

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

Q1. Detection Performance Comparison

Given:

- Model A (Viola-Jones): TP = 80, FP = 20, FN = 30
- Model B (YOLO): TP = 90, FP = 15, FN = 20

(a) Precision and Recall

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Model A

$$\text{Precision} = 80 / (80 + 20)$$

$$= 80 / 100$$

$$= 0.80 = 80\%$$

$$\text{Recall} = 80 / (80 + 30)$$

$$= 80 / 110$$

$$= 0.7273 = 72.73\%$$

Model B

$$\text{Precision} = 90 / (90 + 15)$$

$$= 90 / 105$$

$$= 0.8571 = 85.71\%$$

$$\text{Recall} = 90 / (90 + 20)$$

$$= 90 / 110$$

$$= 0.8182 = 81.82\%$$

(b) Comparison

Model	Precision	Recall
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Viola-Jones	80%	72.73%
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YOLO	85.71%	81.82%
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Conclusion: YOLO is better because it has both higher precision and higher recall. Therefore, YOLO provides better overall object detection performance.

Q2. Motion Estimation Error Analysis

Given errors:

- Method A: 2, 3, 2 pixels
- Method B: 3, 5, 4 pixels

(a) Average Error

Formula:

$$\text{Average Error} = \text{Sum of Errors} / \text{Number of Frames}$$

Method A

$$= (2 + 3 + 2) / 3$$

$$= 7 / 3$$

$$= 2.33 \text{ pixels}$$

Method B

$$= (3 + 5 + 4) / 3$$

$$= 12 / 3$$

$$= 4 \text{ pixels}$$

(b) Comparison

Method	Average Error
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Method A	2.33 pixels
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Method B	4 pixels
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Conclusion: Method A is more accurate because it has a lower average

displacement error. Lower error indicates better motion estimation accuracy.

Q5. Pattern Recognition Accuracy Comparison

Given:

- Traditional system: 85 correct predictions out of 100
- Deep Learning: 92 correct predictions out of 100

(a) Accuracy

Formula:

Accuracy = Correct Predictions / Total Samples × 100

Traditional

= 85 / 100 × 100

= 85%

Deep Learning

= 92 / 100 × 100

= 92%

(b) Comparison

System	Accuracy
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Traditional	85%
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Deep Learning 92%

Conclusion: Deep Learning performs better because its accuracy is 92%, which is 7 percentage points higher than the traditional system.

Q9. Detection Accuracy Comparison

Given:

- Model A: 180 correct out of 200
- Model B: 170 correct out of 200

(a) Accuracy

Formula:

Accuracy = Correct Predictions / Total Predictions × 100

Model A

= 180 / 200 × 100

= 90%

Model B

= 170 / 200 × 100

= 85%

(b) Comparison

Model Accuracy

A	90%
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B	85%
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Conclusion: Model A is better because it achieves 90% accuracy, which is 5 percentage points higher than Model B.

Q10. Error Rate Comparison

Given:

- Method A: 20 errors out of 100
- Method B: 10 errors out of 100

(a) Error Rate

Formula:

Error Rate = Errors / Total Samples × 100

Method A

= 20 / 100 × 100

= 20%

Method B

= 10 / 100 × 100

= 10%

(b) Comparison

Method Error Rate

A 20%

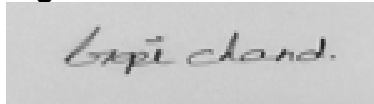
B 10%

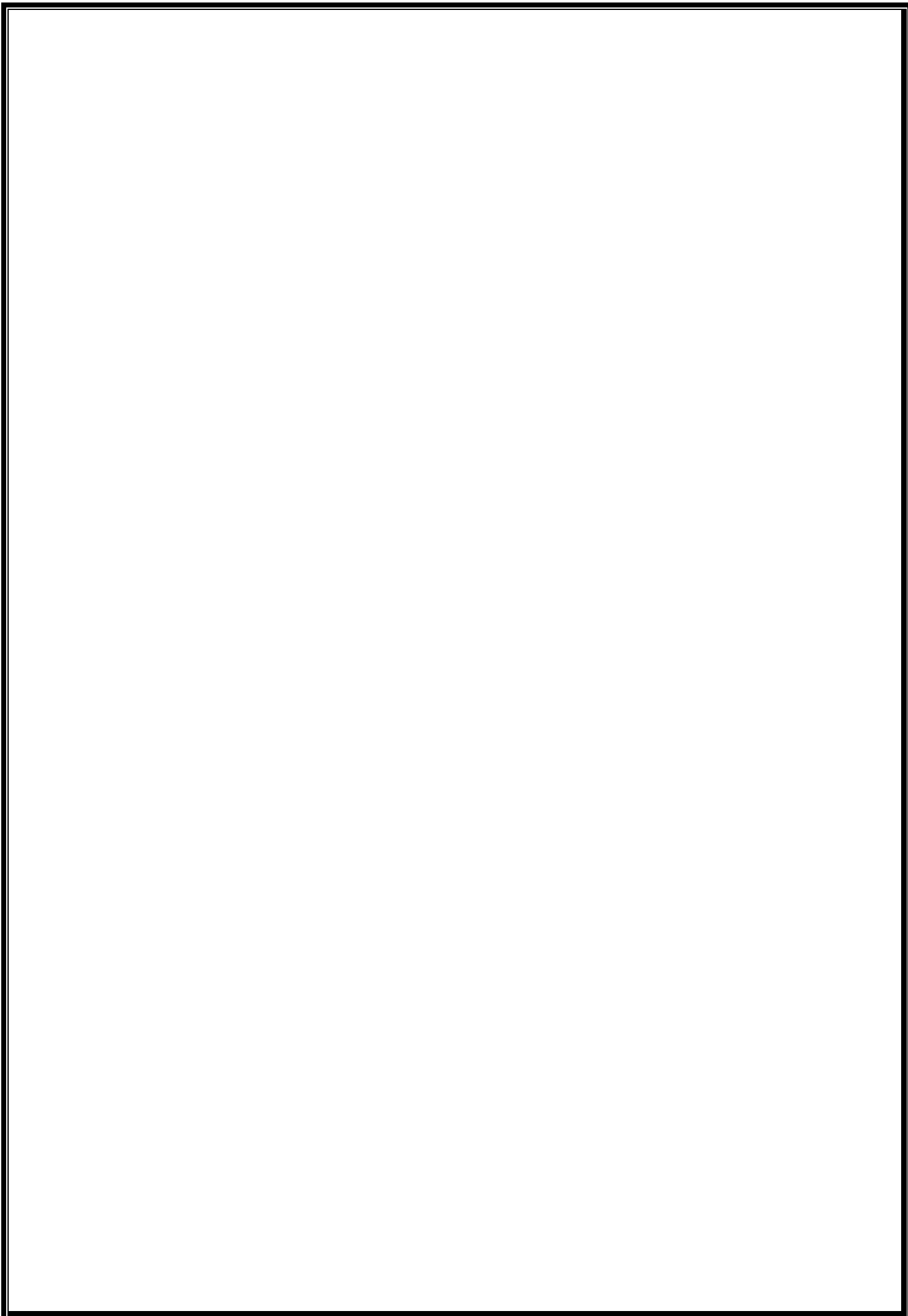
Conclusion: Method B is better because it has a lower error rate of 10% compared with Method A's 20%. Lower error rate indicates better performance.

Code of Conduct:

I G. Gopi Chand (192424384)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in cursive script that reads "Gopi Chand".



RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand